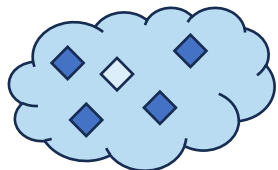


多模态输入特征

文本

...I had a pretty good day today. Everything went more smoothly than I expected....

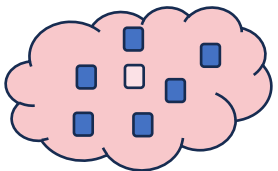
文本特征序列



$$X^T \in \mathbb{R}^{50 \times 768}$$

语音

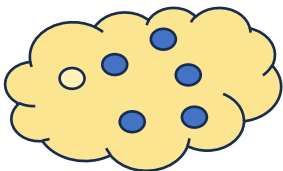
语音特征序列



$$X^A \in \mathbb{R}^{50 \times 74}$$

视觉

视觉特征序列



$$X^V \in \mathbb{R}^{50 \times 35}$$

模态双路池化

$m \in \{T, A, V\}$, 三路独立运行

掩码均值池化

$$\mu_m = (\sum_t p_t x_{m,t}) / (\sum_t p_t)$$

模态映射 $P_m(\mu_m)$

同一模态双路共享 P_m

时间注意力池化

线性标量评分

$$s_{m,t} = w_m^T x_{m,t} + b_m$$

$$p_t = 0 \rightarrow s_{m,t} = -\infty$$

$$\alpha_m = \text{softmax}(s_m)$$

$$a_m = \sum_t \alpha_{m,t} x_{m,t}$$

模态映射 $P_m(a_m)$

$$h_m = P_m(\mu_m) + \gamma_m P_m(a_m)$$

多模态融合

Concat三模态拼接

融合多层感知机

融合表示 z

双任务情感预测

情感极性分类
三分类输出



正向



中性



负向

情感强度回归
连续值输出

-3 0 +3

$\hat{y} = 1.42$